

ASSIGNMENT

Handling Missing Data in ETL

Q1. What are the most common reasons for missing data in ETL pipelines?

ANS

Reason	Explanation	Example
Human/Manual Errors	Mistakes during data entry or incomplete forms	A customer forgets to fill in their income field
System/Process Failures	Issues during extraction or transfer cause data loss	Network interruption while pulling records from a CRM
Inconsistent Source Systems	Different formats or optional fields across sources	One branch records "Region," another leaves it blank
Data Corruption	Errors in files or databases make values unreadable	A corrupted CSV file with missing sales values
Business Rules/Intentional Omission	Certain fields are optional or not collected	A survey that doesn't ask income for some respondents

Q2. Why is blindly deleting rows with missing values considered a bad practice in ETL?

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Issue	Explanation	Impact on Analytics
Loss of Valuable Data	Deleting rows removes not just the missing field but all other valid information in that row.	Customer demographics, sales, or region info may be lost.
Bias in Results	If missingness is systematic (e.g., certain regions or income groups), deletion skews the dataset.	Dashboards may misrepresent trends or KPIs.
Reduced Sample Size	Fewer records mean weaker statistical power and less reliable insights.	Forecasts and hypothesis tests become less accurate.
Missed Business Insights	Sometimes missingness itself carries meaning (e.g., customers not disclosing income).	Could indicate privacy concerns or market behavior.

Q3. Explain the difference between: Listwise deletion Column deletion

ANS 1. Listwise Deletion (Row-Level Removal)

- Removes the entire row when any selected value is missing.
- Use only when missing % is very small (<5%).
- Example: Remove rows where Region is blank.

2. Column Deletion

- Remove the entire column when > 60–70% values are missing.
- Use for non-critical attributes.

Q4. Why is median imputation preferred over mean imputation for skewed data such as income?

ANS Median imputation is preferred over mean imputation for skewed data like income because the mean gets pulled by extreme outliers, while the median stays stable and represents the central tendency better.

- Median Imputation Scenario: Best for skewed data such as income, house prices, or medical costs where a few very high values distort the mean.
- Mean Imputation Scenario: Works well for normally distributed data (e.g., heights, weights, exam scores) where values are evenly spread and outliers are minimal.

Q5. What is forward fill and in what type of dataset is it most useful?

ANS Forward Fill: This technique replaces a missing value with the last available (previous) value in the column.

Example: If daily sales data is missing for June 3, it will take the value from June 2.

Forward fill is mainly used in **time-series datasets** (like stock prices, weather data, or daily sales).

Q6. Why should flagging missing values be done before imputation in an ETL workflow?

ANS Flagging first means marking which records have missing values before filling them.

This step preserves important information: sometimes the fact that a value is missing carries meaning (e.g., customers not disclosing income).

If you jump straight to imputation, you lose the ability to distinguish between original and filled values.

Flagging ensures analysts can later check whether imputed values influenced trends or KPIs.

Q7. Consider a scenario where income is missing for many customers.

How can this missingness itself provide business insights?

ANS When many customers don't provide income, the missingness itself can reveal insights:

- It may show privacy concerns — customers are uncomfortable sharing sensitive details.
- It can highlight patterns in certain groups — for example, high-income or low-income customers may be more likely to skip the field.
- It signals business behavior — customers who hide income might also avoid premium offers or targeted marketing.

SECTION B : PRACTICAL

Q8. Listwise Deletion Remove all rows where Region is missing.

Tasks: Identify affected rows

Show the dataset after deletion

Mention how many records were lost

Q10. Flagging Missing Data Create a flag column for missing Income.

Tasks: Create Income_Missing_Flag (0 = present, 1 = missing)

Show updated dataset

Count how many customers have missing income

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Customer_ID	Name	City	Monthly_Sales	Income	Region	Income_Missing_Flag
101	Rahul Mehta	Mumbai	12000	65000	West	0
102	Anjali Rao	Bengaluru	NaN	NaN	NaN	1
103	Suresh Iyer	Chennai	15000	72000	South	0
104	Neha Singh	Delhi	NaN	NaN	NaN	1
105	Amit Verma	Pune	14000	58000	North	0
106	Karan Shah	Ahmedabad	NaN	NaN	NaN	1
107	Pooja Das	Jaipur	16000	61000	West	0
108	Riya Kapoor	Kolkata	18000	69000	East	0

New Column: Income_Missing_Flag (0 = present, 1 = missing).

Updated Dataset Example:

101 → Income 65000 → Flag = 0

102 → Income NaN → Flag = 1

104 → Income NaN → Flag = 1

106 → Income NaN → Flag = 1

Count of Missing Income: 3 customers have missing income.